Status	Finished
Started	Thursday, 15 October 2026, 8:56 PM
Completed	Thursday, 15 October 2026, 8:59 PM
Duration	3 mins 11 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It makes it easier to select items based on their relationship to the rest of the items
c. It slows down the process
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **2**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It sorts data using a single pass
c. It does not require recursion
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **3**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It allows for faster searching
b. It makes the search operation slower
c. It has no effect on the search operation
d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It sorts data out of the box efficiently
c. It is less efficient than custom sorting algorithms
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Searching
b. All of the above
c. Duplicates
d. Selection

The correct answer is: All of the above

Question **6**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data in a single pass
b. Sorting data sequentially
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Insertion sort
- c. Quick sort
- d. Heap sort

The correct answer is:
Insertion sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Being the most efficient sorting algorithm
- b. Sorting data in a non-sequential manner
- c. Using the divide-and-conquer approach
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is difficult to implement
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **12**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Searching
c. Merging
d. Sorting

The correct answer is:
Complexity

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes code execution slower
- c. Makes data harder to manage
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Finding a specific element in a list
- c. Arranging data in a specific order
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Heap Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Bubble Sort